Internship Assignment: Data Science Intern (React + FastAPI + MySQL)

Objective

The goal of this assignment is to evaluate your ability to build a data-driven dashboard, work with web scraping, and integrate a full-stack application using modern technologies.

Tech Stack (Mandatory)

Frontend: React.js
Backend API: FastAPI (Python)
Database: MySQL

Task Overview

You are required to build a **Business Listings Dashboard** that collects, stores, and visualizes data.

Part 1: Data Collection (Web Scraping)

Scrape **500+ business listings** from any of the following platforms:

* Google Maps
* Justdial
* Sulekha
* Any other business directory websites

Data Fields to Capture:

* Business Name
* Category
* City
* Address
* Phone Number (if available)
* Source (e.g., Google, Justdial, etc.)

Part 2: Database Design

Create a MySQL database and a table:

Table Name:`listing_master`

Suggested Schema:

```sql
id (Primary Key)
business_name
category
city
address
phone
source
created_at
```

Part 3: Backend Development (FastAPI)

Develop APIs using FastAPI:

Required APIs:

1. Insert Listings API**

   * Bulk insert scraped data into MySQL

2. Fetch Dashboard Data APIs

   * City-wise count
   * Category-wise count
   * Source-wise count

Part 4: Frontend Dashboard (React.js)

Create a simple dashboard UI that displays:

Reports:

* City-wise business count
* Category-wise business count
* Source-wise business count

UI Expectations:

* Clean and minimal design

* Use charts (Bar / Pie) using libraries like:

  * Chart.js / Recharts / ApexCharts

📈 Expected Output

A working dashboard that:

* Displays aggregated data visually
* Fetches data from FastAPI
* Is connected to MySQL

🗂 Submission Requirements

Please submit the following:

1. GitHub Repository (Mandatory)

   * Separate folders for frontend and backend

2. Demo Video (3–5 mins) explaining:

   * Scraping approach
   * API working
   * Dashboard functionality

3. Database Dump (.sql file)

4. README.md with:

   * Setup instructions
   * Tech stack used
   * Challenges faced

⏳ Timeline

* Submission Deadline: **3–5 days from assignment date**

☆ Evaluation Criteria

Criteria	Weightage
Functionality	30%
Code Quality	20%
UI/UX	15%
Data Accuracy	15%
API Design	10%
Documentation	10%

⚠ Important Notes

* Avoid scraping in a way that violates website terms of service
* You may use mock/sample data if scraping is blocked (but explain approach)
* Focus on **logic, structure, and clarity** over perfection
